UNITED STATES MARINE CORPS
COMMUNICATION TRAINING BATTALION
MARINE CORPS COMMUNICATION-ELECTRONICS SCHOOL
BOX 788251
TWENTYNINE PALMS, CALIFORNIA 92278-8251

Annex B

DS.02.03

Group Policy

Student Outline

Data Systems Administrator Course
M09CVQ1
Date of Revision: 20230920

APPROVED BY: CTB ACADEMICS DATE: 20230920

Table of Contents

Terminal Learning Objective 3
Enabling Learning Objectives 3
GPO OVERVIEW 4
Types of GPOs 4
Data Security and Group Policy Object 4
Some of these security measures include: 5
Benefits of Group Policy Objects 5
There are several benefits to implementing GPOs in addition to security, including: 5
Limitations of GPOs 6
The limitations of Group Policy Objects include: 6
Processing order of GPOs 6
Examples of GPOs 7
Best practices 8
Some best practices for GPOs include: 8
Group Policy Core Protocol 9
Group Policy Settings 9
Group Policy Objects 10
Group Policy Extensions 11
Group Policy Data Storage 11
Group Policy Administration 12
Group Policy Application 12
Triggering Group Policy Application 13
Group Policy Scope of Management (SOM) 14
Group Policy Management 16
Group Policy Structure 17
10 Common Problems Causing Group Policy to Not Apply 18
Start with the Scope 19
Dive into Delegation 20
Learn Your Links 20
Loopback 22

Terminal Learning Objective

1. Given a command's mission, threat assessment, operational environment, operational plans, virtual environment, and personnel, establish operating systems in accordance with applicable STIGs and local cybersecurity policies in support of the communications plan. (0671-IMPL-1002)

Enabling Learning Objectives

1. Without the aid of reference, identify in writing the characteristics of Group Policy Objects, in accordance with 0671-IMPL-1002

2. Without the aid of reference, create group policies in accordance with 0671-IMPL-1002

3. Without the aid of reference, implement group policies in accordance with 0671-IMPL-1002

GPO OVERVIEW

Microsoft's Group Policy Object (GPO) is a collection of Group Policy settings that defines what a system will look like and how it will behave for a defined group of users.

Microsoft provides a program snap-in that allows you to use the Group Policy Management Console (GPMC). The selections result in a Group Policy Object. The GPO is associated with selected Active Directory containers, such as sites, domains or organizational units (OU). The GPMC allows you to create a GPO that defines registry-based polices, security options, software installation and maintenance options, scripts options and folder redirection options.

Types of GPOs

There are three types of GPOs:

- Local Group Policy Objects. A local Group Policy Object refers to the collection of group policy settings that only apply to the local computer and to the users who log on to that computer. Local GPOs are used when policy settings need to apply to a single Windows computer or user. Local GPOs exist by default on all Windows computers.

- Non-local Group Policy Objects. A non-local group policy object is used when policy settings have to apply to one or more Windows computers or users. Non-local GPOs apply to Windows computers or users once they're linked to Active Directory objects, such as sites, domains or organizational units.

- Starter Group Policy Objects. Introduced in Windows Server 2008, starter GPOs are templates for Group Policy settings. These objects enable an administrator to create and have a pre-configured group of settings that represent a baseline for any future policy to be created.

Data Security and Group Policy Object

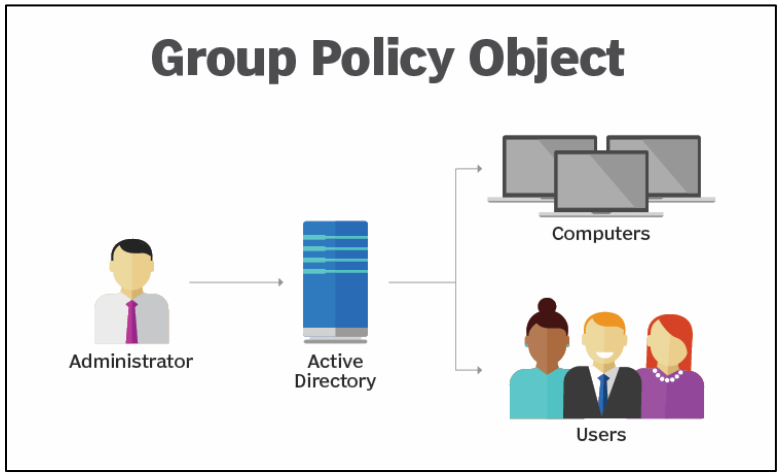
There are some Group Policy settings that can help secure a company's network. For example, through Group Policy, an organization can run scripts, stop users from accessing certain resources and perform simple tasks, such as forcing a particular home page to open for every network user.

Some of these security measures include:

- Limiting access to Control Panel -- through Control Panel, a company can control all aspects of a computer. Limiting who has access to a computer enables organizations to keep data and other resources safe.

- Disabling Command Prompt -- A company can use Command Prompts to run commands that give high-level access to users and bypass other system restrictions. That's why it's prudent to disable Command Prompt to ensure the security of system resources. If a user tries to open a command window after Command Prompt has been disabled, the system will display a message indicating that some settings are preventing this.

- Prevent software installations -- if users are allowed to install software, they may install unwanted applications or malware that can compromise a company's system. As such, it's better to prevent software installations through Group Policy. *This is key among most USMC networks.



Benefits of Group Policy Objects

There are several benefits to implementing GPOs in addition to security, including:

- More efficient management -- GPOs already in place apply a standardized environment to all new users and computers that join an organization's domain, saving time on setup.

- Ease of administration -- system administrators can deploy software, patches and other updates via GPO.

- Better password policy enforcement -- GPOs determine password length, reuse rules and establish other requirements for passwords to keep a company's network safe.

- Configuring folder redirection -- GPOs enable companies to ensure users are keeping important company files on a centralized and monitored storage system. For instance, an organization can redirect a user's Documents folder, which is usually stored on a local drive, to a network location.

Limitations of GPOs

The limitations of Group Policy Objects include:

- They run sequentially -- GPOs process actions one after another. Consequently, if many GPOs have to be configured, it can take a long time for users to log on if improperly configured.

- Flexibility is limited -- GPOs can only be applied to users or computers. So they're limited when it comes to applying settings based on context.

- Limited triggers -- GPOs can only be applied at computer startup, when a user logs on or at set intervals. GPOs can't react to changes in environment, such as network disconnect or reconnect.

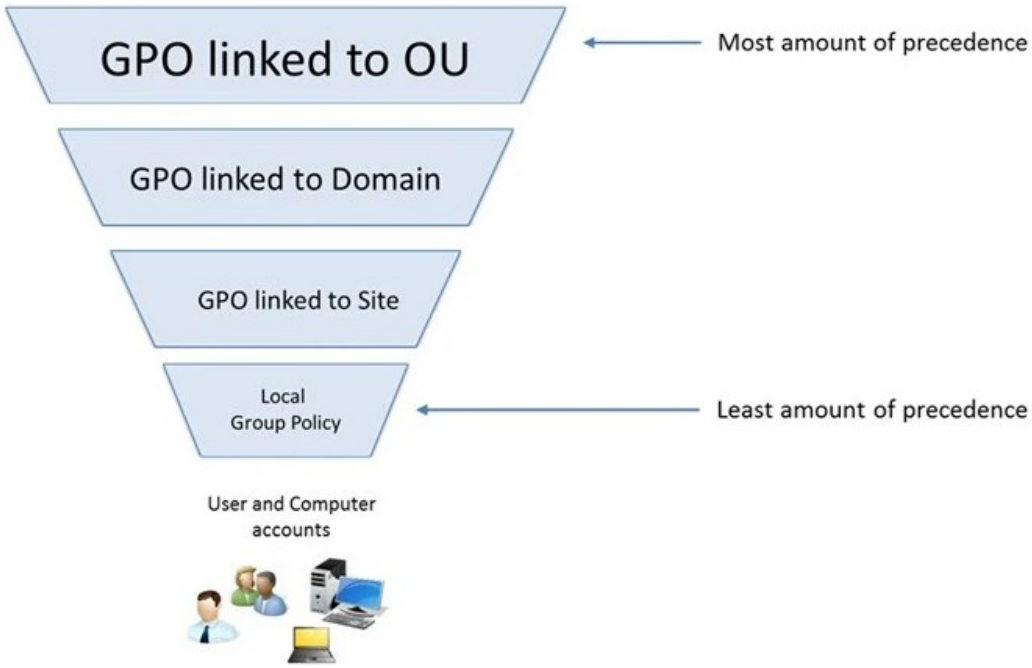
- Difficult to maintain -- there's no built-in search or filter option to find a specific setting within a GPO, making it difficult to find or fix issues with existing settings. Incorrect naming standards can create confusion.

- No Version control -- changes made to GPO settings aren't audited. So if an incorrect change is made, it's impossible to tell what the change was or who made it.

Processing order of GPOs

The processing order of Group Policies effects what settings are applied to the computer or end-user.

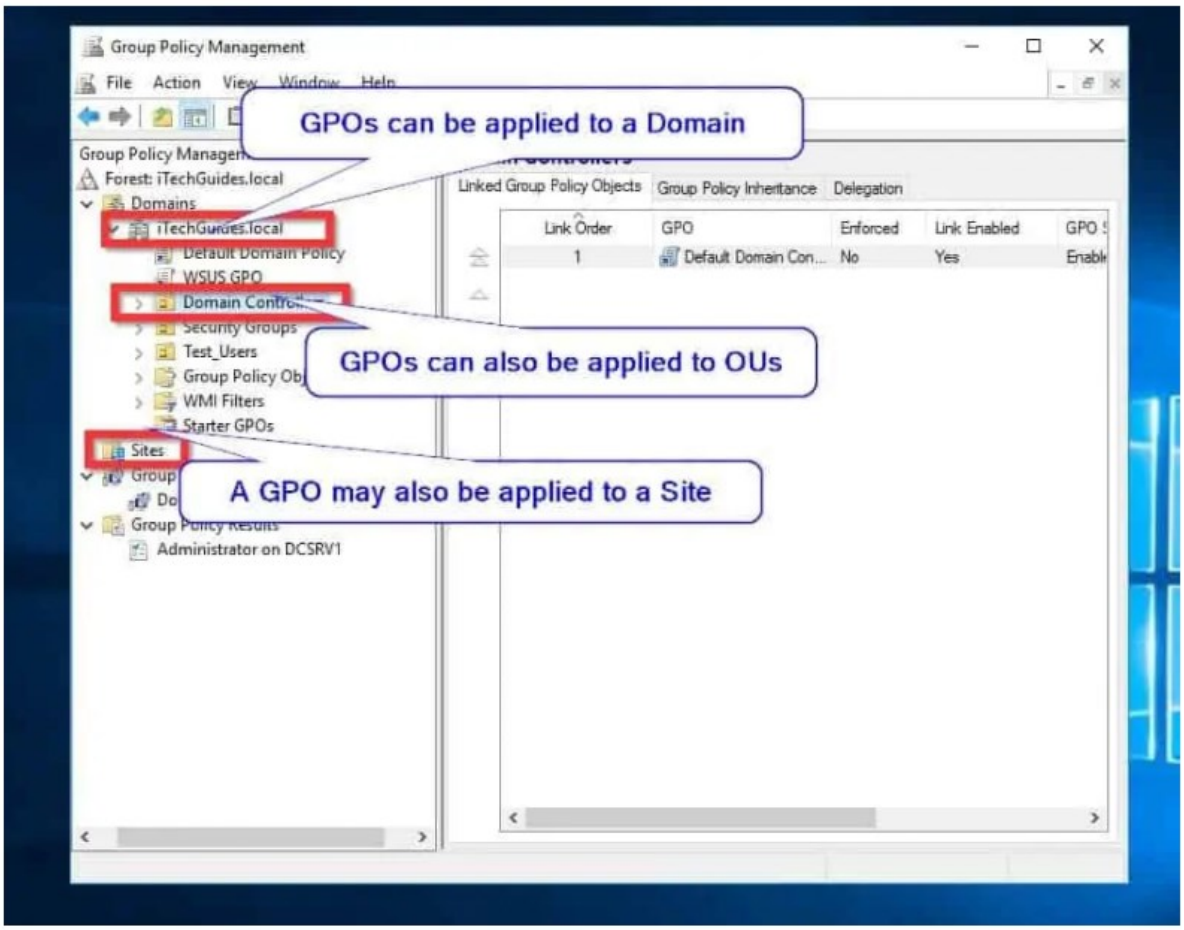
This **processing order** is known as **LSDOU: Local, Site, Domain, Organization Unit**.
First the local computer policy is processed, followed by Active Directory policies from site level to domain, then into OU (GPOs in nested organizational units apply from the OU closest to the root first, and continues from there). If there are any conflicts, the last applied policy will take effect.



Examples of GPOs

The following are examples of Group Policy Objects:

- A GPO might specify the home page that's first displayed when a user launches Internet Explorer. When the user logs on to the domain, that group policy object is retrieved and applied to the configuration of the user's Internet Explorer.
- An organization can deploy shared network printer connections to users from a specific OU of Active Directory by using Group Policy. So when a user logs in to Windows, an assigned network printer will automatically appear in the list of available printers. These can also be applied at the "site" level which can be directed by "sites and services".
- Admins can use a group policy to adjust settings, such as turning off computer displays are a certain period of time, choosing default programs and preventing users from changing Internet connection options.



Best practices

Some best practices for GPOs include:

- Create a well-designed organizational unit structure in Active Directory to simplify applying and troubleshooting Group Policy.
- Give GPOs descriptive names to enable admins to quickly identify what each GPO does.
- Add comments to each GPO explaining why it was created, what its purpose is and what its settings are.
- Don't set GPOs at the domain level because they'll be applied to all computer and user objects. That could cause some settings to be applied to some objects unnecessarily.
- Don't use the root computers or user folders in Active Directory because they're not organizational units and they can't have GPOs linked to them. When a new user or computer object appears in these folders, it should be immediately to the appropriate OU.

- Don’t disable a GPO. Rather, delete the link from an OU instead of disabling the GPO if you don’t want it to be applied. Disabling the GPO will prevent it from being applied entirely on the domain. That could be a problem because if that particular Group Policy is used in another OU, it won’t work there any longer.

Group Policy Core Protocol

The Group Policy: Core Protocol [MS-GPOL] is a client/server protocol that enables a Group Policy client to discover and retrieve policy settings that are created by a Group Policy administrator (a domain administrator) and are stored as a Group Policy Object (GPO) in Active Directory ([MS-ADTS]). A Group Policy administrator creates policy settings to control Group Policy client behavior and capabilities. The Group Policy Core Protocol then facilitates the communication of the administrator-defined policies from the Group Policy server to domain members such as a Group Policy client or a user who is interactively logged on to the Group Policy client computer.

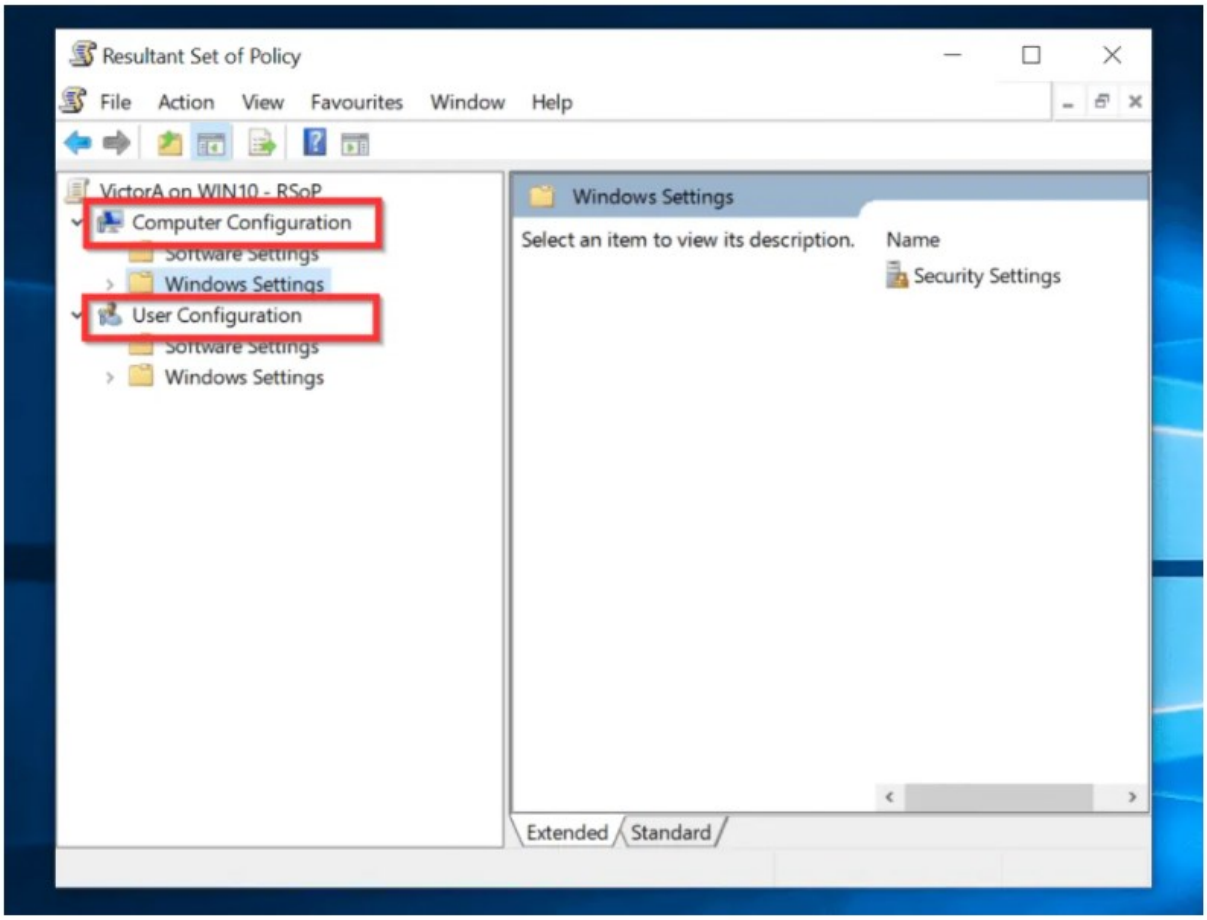
Group Policy Settings

User policy settings: These specify capabilities and behaviors for interactively logged-on users. These settings can also affect different users who are logged on to the same computer. Examples of such settings include the user's default location for saving documents, or the desktop background image for a user.

Some settings affect users regardless of the computer that they log on to.
For example, policy source mode, can override user policy settings by causing computer policy settings to be applied to the user.

Computer policy settings: These specify capabilities and behaviors for individual computers, even when no users are logged on. Computer policy settings can also globally affect every user who logs on to the computer. Examples include policy settings that enable a computer to host a web server, schedule automated disk backups of the computer, or specify a standard web home page for all users of the computer.

The Group Policy Core Protocol enables Group Policy clients to discover and retrieve these policy settings. The policy settings that are applied to the Group Policy client depend on the filtered GPO list, which is derived and prioritized by the core Group Policy engine on the Group Policy client. The filtered GPO list is a set of GPOs that have passed various test criteria to verify whether they are permitted or denied applicability on the Group Policy client.



Group Policy Objects

Group Policy uses several protocols to create, read, update, and remove GPOs. Group Policy uses a document-centric approach to create, store, and associate policy settings. Group Policy settings are contained in GPOs to maintain various sets of behavior specifications. A GPO is a virtual object that stores policy-setting information with two components:

Directory service: GPOs and their attributes are stored in a directory service, such as Active Directory.

File share: GPOs also store policy settings information on a local or remote file share, such as the Group Policy file share. The Group Policy file share repository in Windows is a system volume (SYSVOL) share on the Group Policy server.

Both of these storage components can reside on the Group Policy server. Through the hierarchical modeling of Active Directory, GPOs can be linked to site, domain, and organizational unit (OU) containers to enable policy settings to be applied to target users and computers that are associated with these containers. This infrastructure provides a high degree of flexibility that enables the Group Policy administrator to customize configurations, such as delivering a specific piece of software to specialized users based on their membership in an OU.

A GPO is uniquely identified by a globally unique identifier (GUID). GPO settings are evaluated by the Group Policy client through the hierarchical nature of Active Directory and by interpreting the extension policy file data on the Group Policy file share.

Group Policy Extensions

Group Policy functionality can be enhanced through the implementation of Group Policy extensions. Group Policy extensions consist of client-side extensions (CSEs) and Administrative tool extensions. Most Group Policy extensions have these two extension implementation pairs; a CSE that applies policy settings, and an associated administrative-side extension that plugs into the Administrative tool to define policy settings. Group Policy extensions are invoked by the Administrative tool when creating or updating policy settings. Group Policy extensions are also invoked by the core Group Policy engine when applying policy on a policy target such as a Group Policy client.

Group Policy Data Storage

Active Directory data store: This store is part of AD DS implemented on the Group Policy server and serves as a repository for GPOs. GPOs are maintained in Active Directory as type group Policy Container objects within a Group Policy Objects container and are accessed via the Lightweight Directory Access Protocol (LDAP) calls. A GPO maintains policy configuration settings that apply to policy targets, such as a user that is interactively logged on to a Group Policy client.

Group Policy Administration

Group Policy administration consists of creating new GPOs, deleting GPOs, and editing existing policy settings. In policy administration mode, the Group Policy administrator uses the Administrative tool to locate the Group Policy server and interact with the same Active Directory objects as occurs during policy application by the Group Policy client. However, the Administrative tool does not directly apply policy settings to the Group Policy client. Instead, it only enables the Group Policy administrator to create, update, or delete policy settings, and then update the Group Policy server with those configurations via LDAP. Thereafter, following a Group Policy trigger, the Group Policy client accesses those updated or new objects and associated settings during the policy application process.

Group Policy Application

The policy application process utilizes a pull model when it retrieves Group Policy data to apply to the Group Policy client. For example, when retrieving policy settings, the Group Policy client polls the Group Policy server to check for new policy settings specified by the Group Policy administrator that affect either the client computer itself or a domain user that is interactively logged on to the client computer.
To accommodate these requirements, the application of Group Policy is specified in two modes. The first is computer policy mode, which affects the client computer and all users logging on to the client computer; the second is user policy mode, which only affects the users who log on to the client computer. For user policy mode, the policy target is a domain user account, for which policy settings are retrieved. For computer policy mode, the policy target is a domain computer account, for which policy settings are retrieved.
The application of Group Policy is triggered by specific events, such as a user logon or computer startup.

The following is a conceptual summary of the processes that occur whenever Group Policy is applied. The specified actions of the Group Policy client are carried out by the core Group Policy engine running on the Group Policy client:

DC discovery: The Group Policy client searches for a domain controller (DC) and connects to Active Directory.

DN discovery: The Group Policy client attempts to discover the Distinguished Name (DN) of the policy target, which is used in querying for applicable GPOs.

Domain Scope of Management (SOM) search: The Group Policy client queries the Group Policy server for any GPOs that are linked to the domain, which therefore applies to the Group Policy client policy target account.

Scope of Management (SOM) defines hierarchical levels from which GPOs apply to policy targets; these levels include the domain, site, and organizational unit (OU) levels. For example, a domain SOM search returns the DNs of all GPOs that are linked to the domain container, which holds one or more policy targets to which the GPOs applies.

Site SOM search: The Group Policy client queries the Group Policy server for any GPOs that are linked to the site container, which therefore applies to the Group Policy client policy target account.

GPO search: The Group Policy client queries the collection of GPOs defined by the SOM, to obtain various information sets that include the GPO security descriptor, the GPO file system path, GPO version number, the Globally Unique IDentifier (GUIDs) of extensions that apply to the Group Policy client, and other GPO metadata.

GPO filter evaluation: The Group Policy client processes each GPO to check its functionality version, disabled/enabled status, empty status, and security rights. These checks determine whether the GPO is allowed or denied applicability on the Group Policy client.

WMI filter evaluation: The Group Policy client queries the Group Policy server for any Windows Management Instrumentation (WMI) filters that limit the set of GPOs that are to be used by Group Policy extensions.

Policy change event: The Group Policy client raises a local Policy Change event at the end of policy application to indicate that a policy has changed.

Triggering Group Policy Application

Certain events that occur trigger the application of Group Policy, at which time the core Group Policy engine is invoked to initiate the application process. The following events trigger the application of Group Policy in computer policy mode and user policy mode.

Computer policy mode: The following events trigger the application of Group Policy to the Group Policy client computer:

- Computer startup
- Computer shutdown
- Periodic refresh timer

User policy mode: The following events trigger the application of Group Policy to the user on the Group Policy client computer:

- User logon
- User logoff
- Periodic refresh timer

Note- The periodic refresh timer can be superseded to apply Group Policy at any time by using the `GPUpdate /force` **command in the Command Prompt or PowerShell.**

GPUpdate.exe: An update event can be set via GPUpdate.exe to supersede the periodic refresh timer functionality and to allow policy to be applied at any time.

The application of Group Policy in either computer policy mode or user policy mode involves the application of both Administrative template settings and extension settings. However, before this can occur, it is necessary to discover the domain controller that contains the GPOs that apply to the policy targets

Group Policy Scope of Management (SOM)

The collection of GPOs that apply to a set of policy targets is considered the scope of management (SOM). SOM tells the core Group Policy engine which site-, domain-, or OU-level GPOs apply to a policy target. During policy application, the core Group Policy engine searches for GPOs in the Group Policy Objects container in Active Directory and then determines the SOM by inquiring which site, domain, and OU containers the GPOs are linked to, along with the order of precedence in which they apply to the policy target.

SOM is not an object itself but rather a construct that describes how Group Policy is applied to policy targets from Active Directory hierarchical levels by using GPOs. SOM associates GPOs with policy targets that exist within a site, domain, or OU container object, in accordance with the GPOs that are linked to such objects. This association is established, in order of GPO precedence, within a list of GPO DNs that is contained by the gpLink attribute of the site, domain, or OU container object. For example, there might be GPOs at the domain and OU level that apply to a particular set of policy targets, and the order of precedence might be that the OU-level GPO overrides a GPO at the domain-level in terms of certain policy settings that have priority. The GPO applicability and precedence configuration is resolved through various filtering evaluations that result in a final computed list of GPOs whose settings are applied to one or more policy targets.

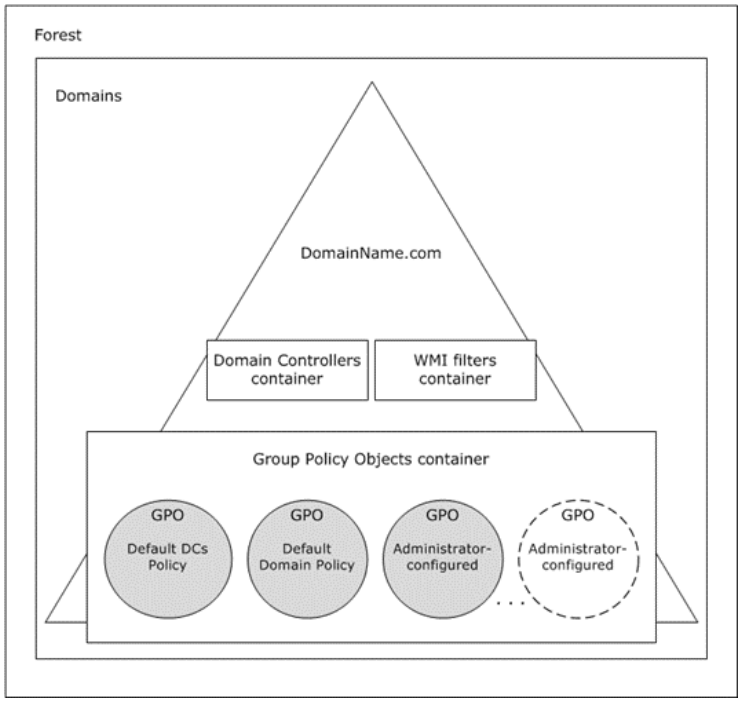
All SOM containers have to maintain the following attributes:

- **SOM DN:** The DN of the SOM container, such as a domain container.
- **gpLink:** A directory string value for the gpLink attribute of the SOM container.
- **gpOptions:** An integer value that is used to set the Group Policy inheritance Configuration among hierarchical SOM containers.
- **SOM object type:** Specifies the type of Active Directory container that the SOM represents; one of the following values is assigned to this attribute:
- **GPLinkOrganizationalUnit:** The SOM container object represents an OU.
- **GPLinkDomain:** The SOM container object represents a domain.
- **GPLinkSite:** The SOM container object represents a site.

NOTE - An Active Directory container comes into the scope of management when one or more GPOs are linked to it.

Group Policy Management

Group Policy can be managed from an interface such as the GPMC, a custom application, or a command-line tool. GPOs exist within a Group Policy Objects container in Active Directory, as shown in the following diagram, and can be managed by a Group Policy administrator:



The Group Policy administrator uses the Active Directory container objects for the domain as shown in the diagram to manage Group Policy. When Group Policy administrators need to manage GPOs, they can **create** a new GPO, **delete** a GPO, or **edit** an existing one. They can also manage policy settings via other default GPOs for the domain. The following default objects and containers can be accessed in a domain for management purposes:

Domain Controllers container: A default container that is automatically created when a server is promoted to a domain controller. It is linked to the domain controller's OU and manages security settings for all domain controllers in a domain.

WMI Filters container: A default container that is automatically created when a server is promoted to a domain controller. It holds WMI filter objects that the Group Policy administrator creates and that are linked to GPOs to exempt specific Group Policy clients from the extension policy settings that they hold.

Group Policy Objects container: A default container that is automatically created when a server is promoted to a domain controller. It provides a hierarchical repository for GPOs that the Group Policy administrator creates with the use of the Administrative tool.

Default Domain Controllers Policy: A default GPO that is automatically created and linked to the domain whenever a server is promoted to a domain controller. This GPO represents the default policy that is applied to all domain controllers in the Domain Controllers container.

Default Domain Policy: A default GPO that is automatically created and linked to the domain whenever a server is promoted to a domain controller. It has the highest precedence of all GPOs linked to the domain, and it applies to all users and computers in the domain. The Default Domain Policy GPO is generally used to manage default account settings, although there are exceptions to this practice. For other areas of policy management, new GPOs can be created; however, some policy settings are best configured at the domain level, and there are no restrictions against doing so.

Administrator-configured: A GPO that is created by the Group Policy administrator to generate custom Group Policy settings for policy targets such as a Group Policy client computer.

Group Policy Structure

Group Policy structure is modeled after the Active Directory structure, in that it has both physical and logical components. At the core of Active Directory's physical architecture is an extensible storage engine that reads and writes information to the Active Directory data store. This engine makes use of the logical, object-based hierarchy that represents data store information.

Group Policy structure is similar to that of **Active Directory**, because it maintains both a logical and physical representation of GPOs, as follows:

Logical component: Consists of a Group Policy container object, which is stored in the Group Policy Objects container of Active Directory. The Group Policy container object contains attributes that specify basic GPO information, such as the following:

- GPO display name
- GPO path to the extension policy and Group Policy template (GPT) files.
- GPO version number
- GPO status
- Access control list (ACL)

- GUID-references to the CSEs that are to be invoked when the core Group Policy engine on the Group Policy client processes the GPO.

When the Group Policy administrator creates a GPO, Active Directory creates a Group Policy container object for that GPO. This Group Policy container is a container object of the group Policy Container class and is named with a GUID that identifies the GPO. The Group Policy container is stored under the `CN=Policies,CN=System` container within the domain. The Administrative tool and the Group Policy client locate this container according to its DN, which is the exact path to the Group Policy container object in the Active Directory data store.

Physical component: Consists of the Group Policy file share component that stores GPT and Group Policy extension settings on a domain controller or other server.
The physical component of a GPO is represented through a series of files containing Administrative template and extension policy settings that are stored on disk. These files contain numerous policy settings along with the state of these settings. These files are stored in Machine and User subdirectories along with the associated GPO version file gpt.ini, in the following path, which is also known as the GPO path:

```
<dns domain name>\<Group Policy file share-name>\<dns domain
name>\Policies\<guid>\
```

Whenever the Group Policy administrator creates a new GPO, the <guid> folder in this path is automatically created and named with the GUID of the GPO. Within the <guid> folder are Machine and User subdirectories that contain extension policy settings and Administrative template configuration items. During policy administration, when the Group Policy administrator creates or modifies Group Policy extension or Administrative template settings, the Administrative tool locates the policy files according to the <guid> in the GPO path. During policy application, the Group Policy client locates the policy files in the same manner.

10 Common Problems Causing Group Policy to Not Apply

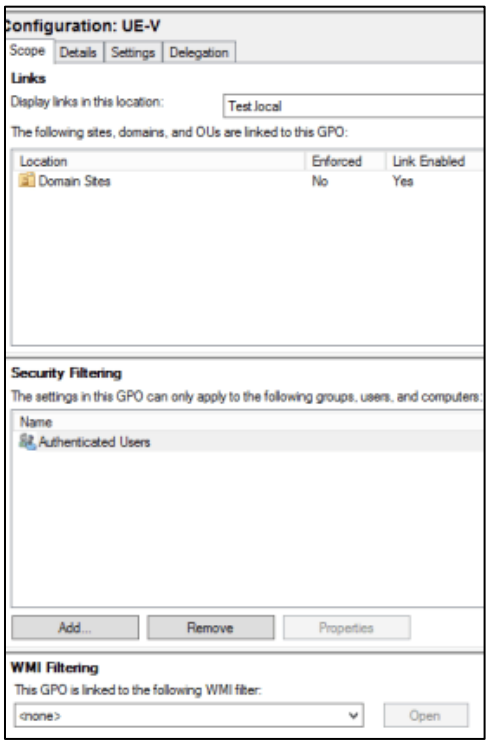
Group Policy is a solid tool and is very stable. Microsoft has made constant improvements to it since Windows 2000. It allows for the configuration and deployment of pretty much anything in your Active Directory environment. From deploying software to setting the default printer Jump, it works. But when it doesn't, Microsoft has provided great guidelines and tools in order to troubleshoot. If Group Policy is not being applied, we can fix it. Let's look at the top ten issues that can stop Group Policy from being applied.

Start with the Scope

1. The most common issue seen with Group Policy is a setting not being applied. The first place to check is the Scope Tab on the Group Policy Object (GPO). If administrators are configuring a computer side setting, make sure the GPO is linked to the Organization Unit (OU) that contains the computer. If the GPO configures a user side setting, it needs to be linked to the OU containing the correct user. Remember, GPOs cannot be linked to an OU that just contains security groups.

2. Next, check the security filtering. Make sure that the computers or users needing the policy are in a group that is specified here. Remember that domain users includes all users, domain computers includes all computer, and authenticated users includes both users and computer. By default, a GPO will be scoped to Authenticated Users.

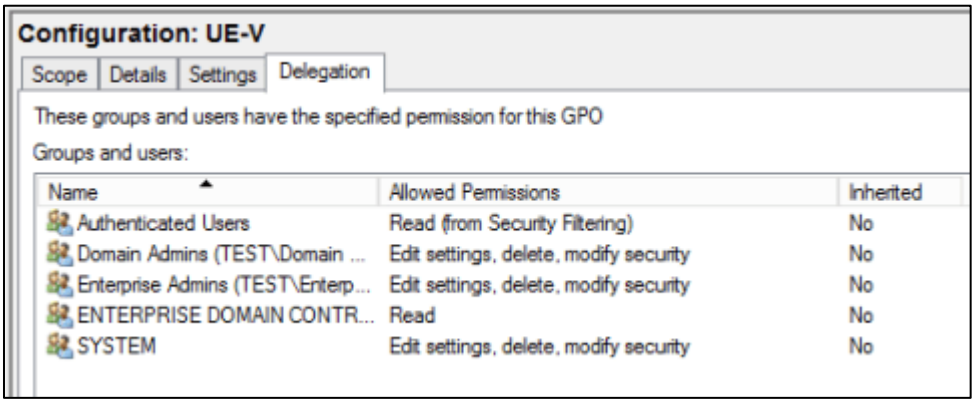
3. Some GPOs make use of WMI filters. These filters can dynamically apply GPOs based on a host of factors. You want a GPO to apply if a device is attached, use WMI. However, that WMI filter has to evaluate to True for the object processing the GPO. This means that if you have a WMI checking a user only setting, you can't scope your GPO only to computers. You can use the WMI validator Jump to check the status of a WMI filter.



The screenshot above recaps the first three common issues. This GPO is linked to an OU named Domain Sites, applies to Authenticated Users, and doesn't have a WMI Filter linked to it. This GPO, which contains several computer side settings, will apply to any computer in the Domain Sites OU.

Dive into Delegation

4. In order for a GPO to apply, the object (a user or a computer) has to have two GPO permissions. It must have Read and Apply Group Policy. By default, an object added to the scope tab receives both of these permissions. Things can get tricky if you are using Deny Permissions to explicitly exclude certain object. Deny permission on the delegation tab would take precedence over any allow.

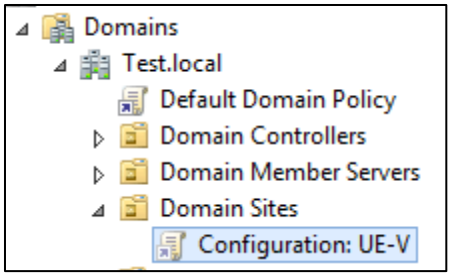


This GPO does not have any Deny permissions set (which show as Advanced settings). If we wanted to exclude a specific group, we could do that here.

Learn Your Links

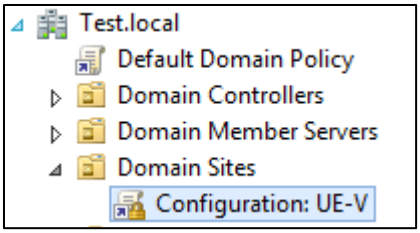
5. GPOs process in a very specific order. The acronym, **LSDOU**, shows that Local GPOs apply first. This is followed by Site, Domain, and finally OU GPOs. In a nutshell, the GPO closest to the object applies last. If you have a GPO linked at the domain that enables Offline Files and a Junior Admin disabled Offline files at the OU level, his GPO wins.

6. When a GPO is created, it lives in the Group Policy Objects container. When you link a GPO to an OU, you are merely creating a shortcut. These links can be enabled or disabled very easily. In the picture below, the Configuration GPO link is disabled. Notice how the link arrow is greyed instead of black (like the Default Domain Policy).

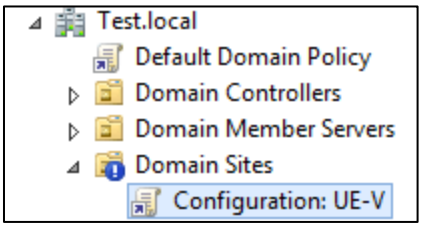


A GPO can be linked to many OUs. These links can be disabled for some OUs and enabled for others. Do not assume that a linked GPO is an enabled GPO.

7. GPOs can also be set to Enforced. An Enforced GPO appears with a lock on the link icon. A GPO upstream (one linked to a higher OU or the domain) that is enforced can cause you problems. If the Default Domain Policy was enforced, every setting in it would apply to every object in the domain. This is because it is linked at the Domain level **(remember LSDOU?)** It does not matter if another GPO is linked an OU and is enforced. With enforcement, the highest GPO wins.



8. The final piece of trickery with Links is the Block Inheritance setting. When an OU is set to Block Inheritance, all GPOs linked above that OU are ignored. In the example below, the Domain Sites OU will not process the Default Domain Policy.



The only exception to this are enforced GPOs. They bust through that Block Inheritance!

Loopback

9. When a computer first starts up, it will process all computer side policies that are linked to the computer's OU (and above). When a user logs on, any user side settings will process that are linked to the user's OU (and above). When loopback is enabled, this process has one more additional step. After the user side items process, any user side settings linked to the computer's OU (and above) are also applied.

Although this does slow down Group Policy Processing, I still love it and find it insanely helpful! With Loopback, I can take a User Side Setting (like setting the homepage in IE) and apply it to a group of computers (such as those in a lab)! Bear in mind that loopback now requires both the User and Computer objects to be added to the scope tab on the GPO. Before Windows Vista, the computer did not need to read permission Jump for the GPO.
Read Carefully

10. Finally, make sure that the GPO is doing what you intend for it to do. When a setting says "Enable Turn Off Audio Mode", it is very easy to get confused. Read carefully over any GPO descriptions when configuring your GPO. You can use Microsoft's GPSearch utility for explanations on GP Settings.